Sentinel

Adaptive time-series anomaly detection

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Challenge 2

Two statistical checks flag unusual events as readings arrive. Each alert includes a short explanation.

Event-F1 0.691

2.8 µs per point

Problem and motivation

- A time series is a sequence of readings, such as CPU usage, traffic, or sales.
- Sentinel must flag unusual behaviour **as each reading arrives**, using only earlier readings.
- Missed incidents matter. Too many false alerts waste people's attention.
- Each result has a score, a NORMAL or ANOMALOUS label, and an explanation.

Goal: accurate event alerts with low delay and low runtime.

Data and preparation

NAB category	Files	Anomaly windows
Artificial	11	6
AWS CloudWatch	17	30
Ad Exchange	6	14
Known Cause	7	19
Traffic	7	14
Tweets	10	33
Total	**58**	**116**

365,558
readings

Learning period

First 15% of each file, up to 750 readings. No alerts or scoring during this period.

Each file stays in its original order; repeated or out-of-order timestamps are handled.

System design

1	**Reading arrives**	Time and value
2	**Check input**	Clean invalid readings
3	**Two detectors**	RobustZ + LaggedResidual
4	**Decide**	Adaptive thresholds + new high/low
5	**Manage event**	Confirm, join, space alerts
6	**Return result**	Score, label, explanation

Every decision uses the current and earlier readings only.

Core algorithm

Part	What it does
RobustZ	Checks whether a reading is far from its usual level, allowing for normal variation.
LaggedResidual	Compares it with the average of 8 readings from roughly 256 points earlier.
Adaptive thresholds	Each detector sets its own alarm level using high learning-period scores.
New high / low	Flags a value beyond the range seen so far by the chosen margin.

RobustZ starts with the median of learning-period readings. It also measures the usual spread, so being above the median alone does not trigger an alert.

LaggedResidual helps detect changes that continue for a while.

Experiment history and tuning

We measured each change before choosing one global default.

1. Compared simple baselines and detector signals on NAB.
2. Searched detector combinations, lag lengths, and window lengths.
3. Tested the new-high / new-low margin and event grouping rules.
4. Selected RobustZ + LaggedResidual, an 8-reading older average, a 256-reading lag, and a 5.75% margin.

Validation: rotate which NAB category is held out when selecting settings.

Evaluation method

An alert inside a labelled anomaly window counts as a true positive. An alert outside all windows is a false positive. A window without an alert is a false negative.

Current default	Value	Across 58 files	Count
Precision	0.731	True positives	76
Recall	0.655	False positives	28
Event-F1	**0.691**	**False negatives**	**40**
Official NAB score	58.80	Median delay	100 readings

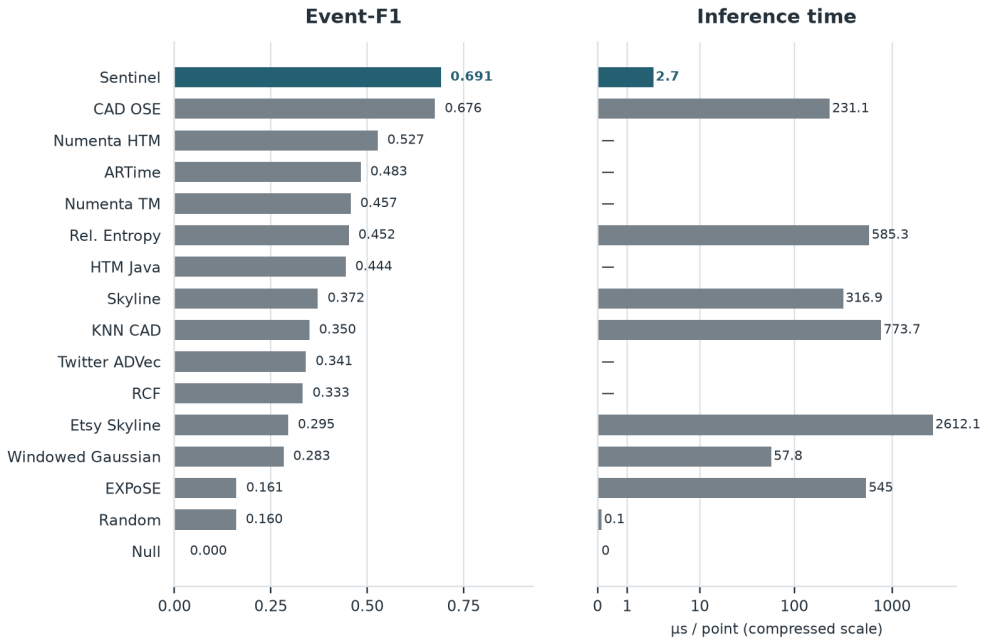
Delay is measured from window start to the first alert, among detected windows only.

Model	Windows found / 116	Median position
Sentinel	**76**	**49.7%**
CAD OSE	92	49.7%
Numenta HTM	96	49.6%
ARTime	106	49.9%
Windowed Gaussian	73	48.8%

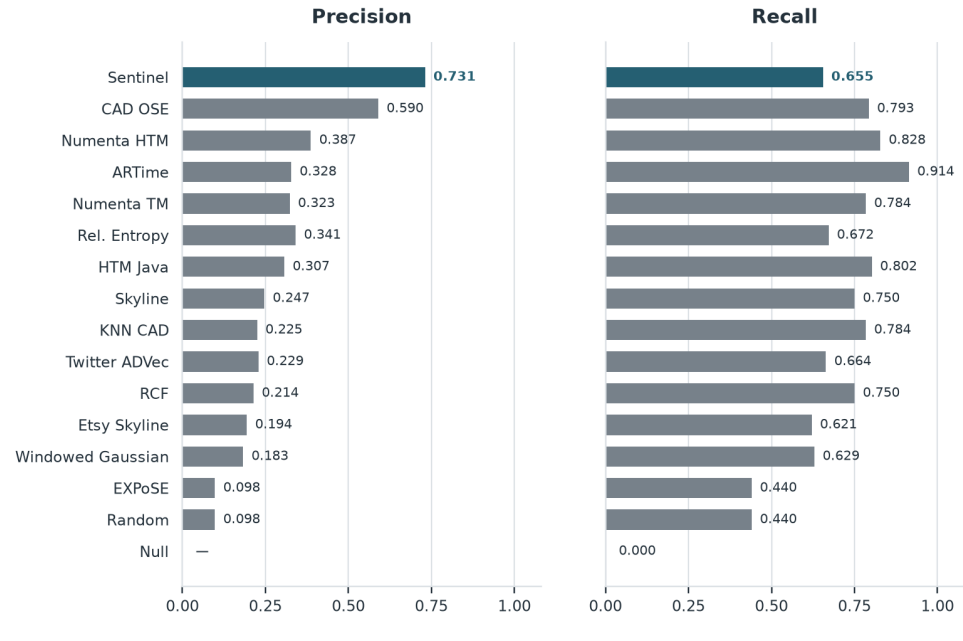
Comparison with other models

Sentinel default: F1 0.691 | precision 0.731 | recall 0.655 | 2.7 µs per point

Event-F1 and inference time by model

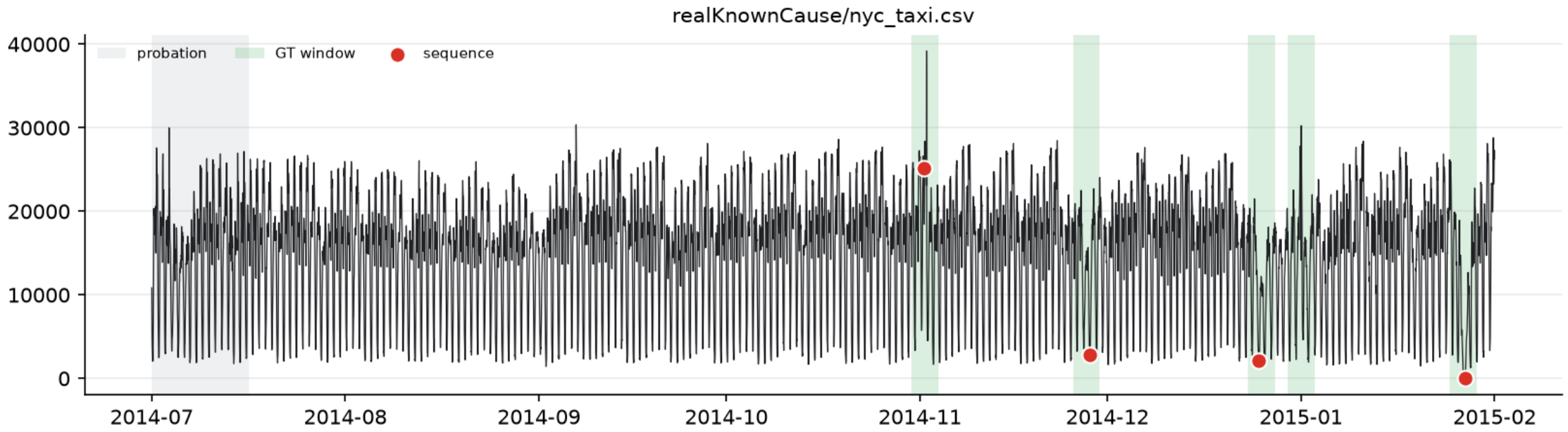


Precision and recall by model



Left: event-F1 and measured runtime. Right: precision and recall. Runtime axis is compressed; labels show actual µs. A dash means unavailable. Published models use NAB's stored scores and thresholds.

Example alert and evidence



NYC taxi, 1 November 2014

A reading rose to 25,115, versus an older-window average near 2,602. LaggedResidual flagged the difference.

Green bands: labelled windows. Red dots: event starts. Sentinel finds 4 of 5 windows here, with 0 false events.

Limits and next steps

- **40 of 116** labelled windows are missed. Changes in daily patterns can be hard to spot when readings remain within the usual range.
- The default was selected and measured on public NAB data. Performance on hidden data is unknown.
- Next: detect pattern changes while keeping false alerts and runtime low.

Conclusion

Two causal statistical checks produce clear alerts at 0.691 event-F1 and about 2.8 µs per point.